

ARJUN TANDON

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Summary

Software Engineer with 3.9 years of experience designing and operating production-grade backend systems and AI platforms. Proven Track record architecting multi-service async orchestration layers, streaming APIs, and high-throughput query processing backends - including LLM-powered platforms serving 30,000+ users. Strong grasp in problem solving, system design, API design, service orchestration, distributed state management, and database engineering.

Experience

Infosys

July 2022 - Present

Digital Specialist Engineer L1 (Apr 2025 - Present)

Bengaluru, Karnataka

- Migrated the natural language query platform from SQL to DAX to support complex multidimensional analytical queries on enterprise datasets.
- Improved DAX query accuracy from 60% to 90% by applying Chain-of-Thought prompting and agentic reasoning principles - designing a multi-step query decomposition approach that restructured how ambiguous, multi-part analytical requests were resolved.
- Designed multi-agent orchestration pipelines using LangGraph - composing nested agent graphs where a parent agent delegated subtasks to specialised child agent graphs, enabling modular analytical reasoning workflows.
- Built a conversational data analysis agent POC - a stateful, multi-turn LLM-powered assistant capable of iterative analytical reasoning over enterprise data, demonstrating agentic patterns ahead of broader adoption.
- Implemented RAG using Elasticsearch-backed semantic retrieval to replace full schema injection, reducing LLM payload size and improving query resolution latency for large enterprise datasets.
- Re-architected a single synchronous endpoint into a multi-service async orchestration layer with persistent session state, enabling complex iterative analytical workflows with stateful LLM interactions.
- Migrated REST endpoints to streaming APIs for long-running LLM inference, reducing client-side timeouts and improving perceived responsiveness.

Infosys

Digital Specialist Engineer L0 (Jul 2022 - Apr 2025)

Bengaluru, Karnataka

- Solely designed and built an internal Text-to-SQL natural language query platform from scratch, serving 100+ finance users at 95% query accuracy via iterative validation, prompt engineering, and business rule enforcement pipelines.
- Key contributor to Navi Data Assist, a company-wide natural language data access platform that scaled to 30,000+ users; drove architectural decisions, resolved key blockers, and led technical problem-solving.
- Architected the LLM query processing backend: natural language to validated SQL via prompt engineering, and structured results with enforced schema constraints.
- Owned end-to-end AI pipeline performance optimisation: prompt payload reduction, selective schema injection, LLM execution environment controls, and output validation.

Projects

SyncLabs - Realtime Collaborative Code Editor | [GitHub](#)

- Built a fullstack realtime collaborative coding platform supporting simultaneous multi-user editing with live state synchronisation across distributed clients, synchronised file explorer, live cursors, presence, and commenting.
- Implemented CRDT-based conflict resolution using Y.js and Liveblocks, ensuring consistency and low-latency updates without centralised locking. Designed role-based access control (owner, admin, editor, viewer) for room level access and sharing.
- Used NextAuth for app level auth, Liveblocks id tokens for room level access and upstash redis for rate limiting.

Awards

IS RISE Award - Best AI Project

Infosys, H1-FY'24

Awarded Certificate of Recognition for outstanding achievement in the category Best AI Project at the IS RISE Awards H1-FY'24.

Insta Award - Navi Data Assist

Infosys, Dec 2023

Awarded Certificate of Appreciation under the Infosys Insta Awards for remarkable contributions to the Navi Data Assist project.

Technical Skills

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, Flask, Node.js, REST APIs, Streaming APIs, Async Orchestration, Langchain, LangGraph, Express

Data & Infra: PostgreSQL, MongoDB, MSSQL, Redis, Elasticsearch, Docker, Git, Github, CI/CD, Cloud AI Services

Frontend: React, Next.js, tailwindcss, gsap

AI / ML: AI Pipelines, NLP, Chatbot Development, Semantic Retrieval, Output Validation

GenAI & LLM: Large Language Models, Prompt Engineering, Chain-of-Thought, prompt chaining, LLM Orchestration, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Agentic Reasoning, Streaming Inference

Education

Indian Institute of Information Technology, Kalyani

2018 - 2022

Bachelor of Technology, Computer Science and Engineering

CGPA: 8.32/10